bcs_paper_reproduction_executed

BCS quantum simulation: paper reproduction¶

Notebook reprodutível para acompanhar o manuscrito `papers/bcs_quantum_simulation.md`.

Ele assume que os experimentos já foram executados e que as tabelas/figuras existem em `results/`. A última seção traz os comandos para regenerar tudo.

In [1]:

```
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import Image, display, Markdown

cwd = Path.cwd().resolve()
ROOT = cwd if (cwd / 'results').exists() else cwd.parent
TABLES = ROOT / 'results' / 'tables'
FIGURES = ROOT / 'results' / 'figures'

pd.set_option('display.max_columns', 80)
pd.set_option('display.precision', 6)
```

Parâmetros principais¶

O caso padrão usado no paper é o Hamiltoniano BCS reduzido com quatro modos de pares.

In [2]:

```
params = {
    'xi': [-1.0, -0.35, 0.35, 1.0],
    'g': 0.8,
    'occupied': [0, 1],
    'time': 2.5,
    'final_time_observables': 6.0,
    'trotter_steps': 8,
    'trotter_order': 2,
}
```

```
params
```

Out[2]:

```
{'xi': [-1.0, -0.35, 0.35, 1.0],
 'g': 0.8,
 'occupied': [0, 1],
 'time': 2.5,
 'final_time_observables': 6.0,
 'trotter_steps': 8,
 'trotter_order': 2}
```

Funções auxiliares¶

In [3]:

```
def read_table(name):
    path = TABLES / name
    if not path.exists():
        raise FileNotFoundError(f'Missing table: {path}')
    return pd.read_csv(path)

def show_figure(name, width=850):
    path = FIGURES / name
    if not path.exists():
        raise FileNotFoundError(f'Missing figure: {path}')
    display(Image(filename=str(path), width=width))

def section(title):
    display(Markdown(f'## {title}'))
```

Dinâmica BCS e erro de Trotter¶

In [4]:

```
trotter = read_table('bcs_dynamics_trotter.csv')
display(trotter)
best = trotter.sort_values('fidelity', ascending=False).iloc[0]
best[['time', 'steps', 'order', 'fidelity', 'state_error', 'exact_gap_proxy', 'trotter_gap_p
```

	time	steps	order	fidelity	state_error	exact_energy	trotter_energy	exact_pair_number
0	2.5	1.0	1.0	0.016402	1.459181	2.7	-2.723483	2.0
1	2.5	2.0	1.0	0.422325	1.720592	2.7	0.591860	2.0
2	2.5	4.0	1.0	0.331926	0.964730	2.7	1.138241	2.0
3	2.5	8.0	1.0	0.905103	0.313054	2.7	2.725392	2.0
4	2.5	16.0	1.0	0.983735	0.129330	2.7	2.758153	2.0
5	2.5	32.0	1.0	0.996618	0.061080	2.7	2.736274	2.0

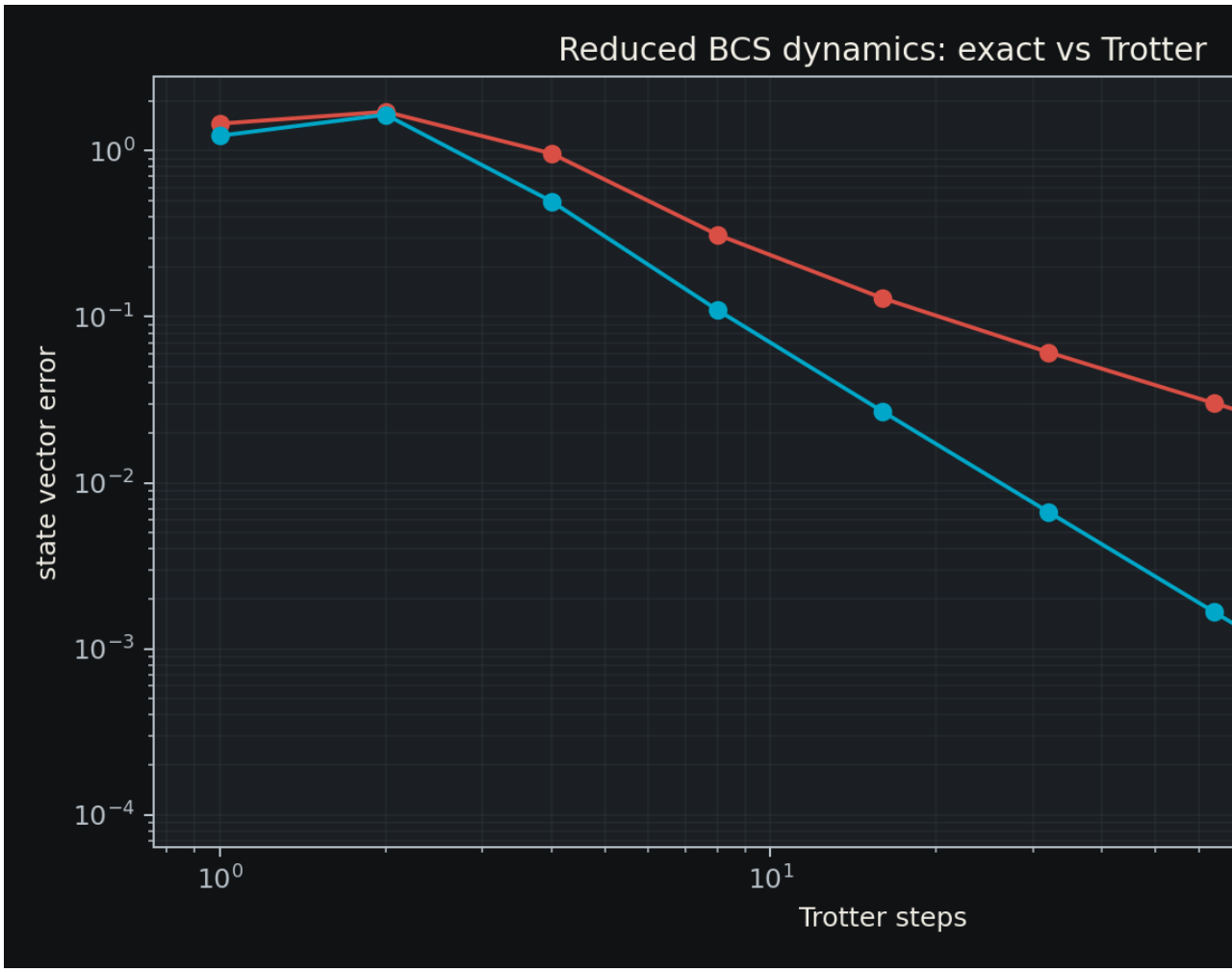
	time	steps	order	fidelity	state_error	exact_energy	trotter_energy	exact_pair_number
6	2.5	64.0	1.0	0.999226	0.030118	2.7	2.719874	2.0
7	2.5	128.0	1.0	0.999814	0.015015	2.7	2.710382	2.0
8	2.5	256.0	1.0	0.999955	0.007505	2.7	2.705305	2.0
9	2.5	1.0	2.0	0.684071	1.233649	2.7	2.154363	2.0
10	2.5	2.0	2.0	0.144306	1.652012	2.7	-0.881944	2.0
11	2.5	4.0	2.0	0.858119	0.496573	2.7	1.926304	2.0
12	2.5	8.0	2.0	0.994675	0.110136	2.7	2.611660	2.0
13	2.5	16.0	2.0	0.999706	0.026798	2.7	2.681008	2.0
14	2.5	32.0	2.0	0.999982	0.006655	2.7	2.695413	2.0
15	2.5	64.0	2.0	0.999999	0.001661	2.7	2.698863	2.0
16	2.5	128.0	2.0	1.000000	0.000415	2.7	2.699716	2.0
17	2.5	256.0	2.0	1.000000	0.000104	2.7	2.699929	2.0

Out[4]:

```
time                   2.500000
steps                256.000000
order                  2.000000
fidelity               1.000000
state_error            0.000104
exact_gap_proxy        0.803355
trotter_gap_proxy      0.803360
Name: 17, dtype: float64
```

In [5]:

```
show_figure('bcs_trotter_error.png')
```



Observáveis físicos para o paper¶

In [6]:

```
obs = read_table('bcs_paper_observables.csv')
final_obs = obs.iloc[-1]
final_obs[['time', 'fidelity', 'state_error', 'exact_pair_number', 'trotter_pair_number', 'e
```

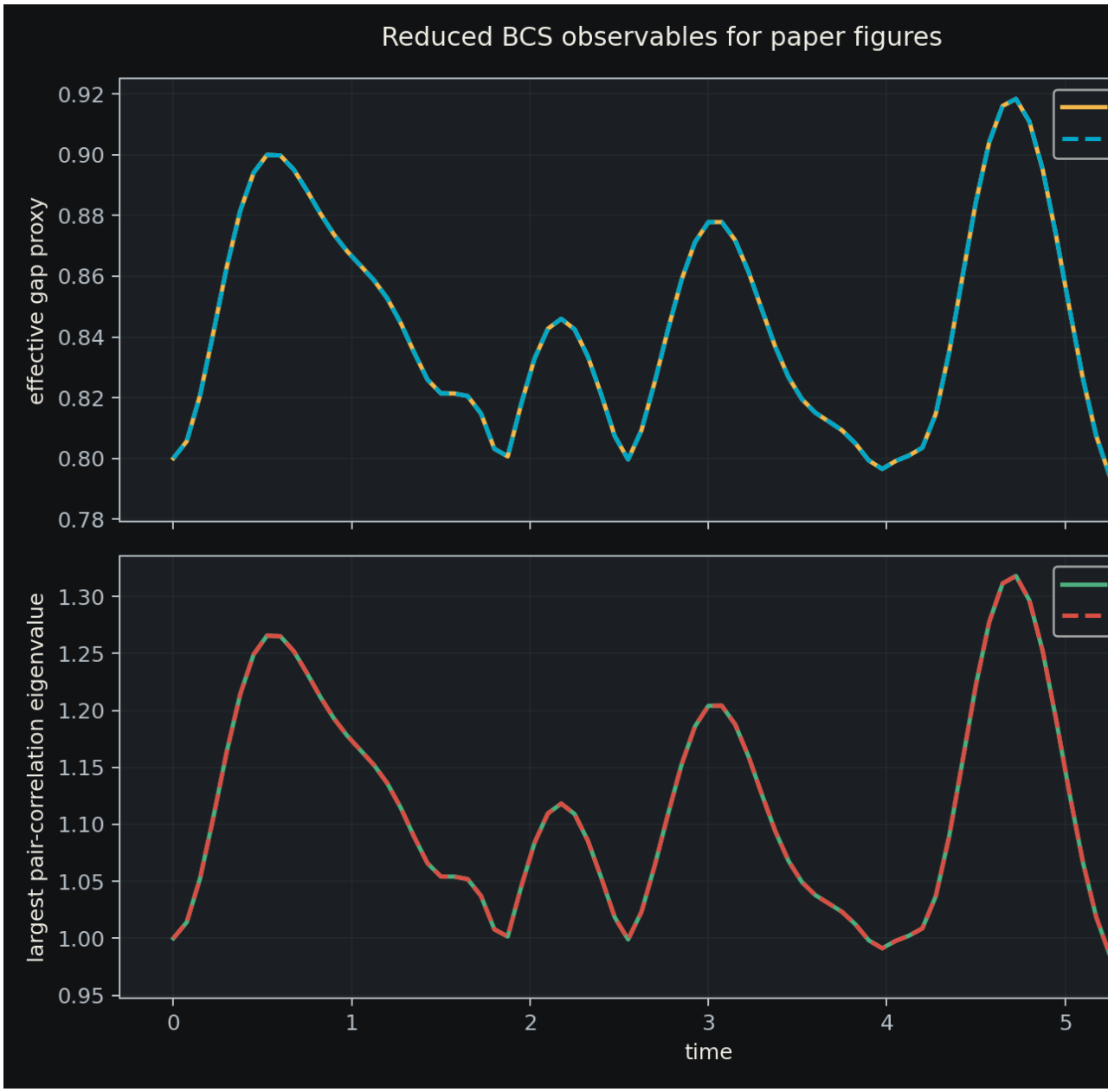
Out[6]:

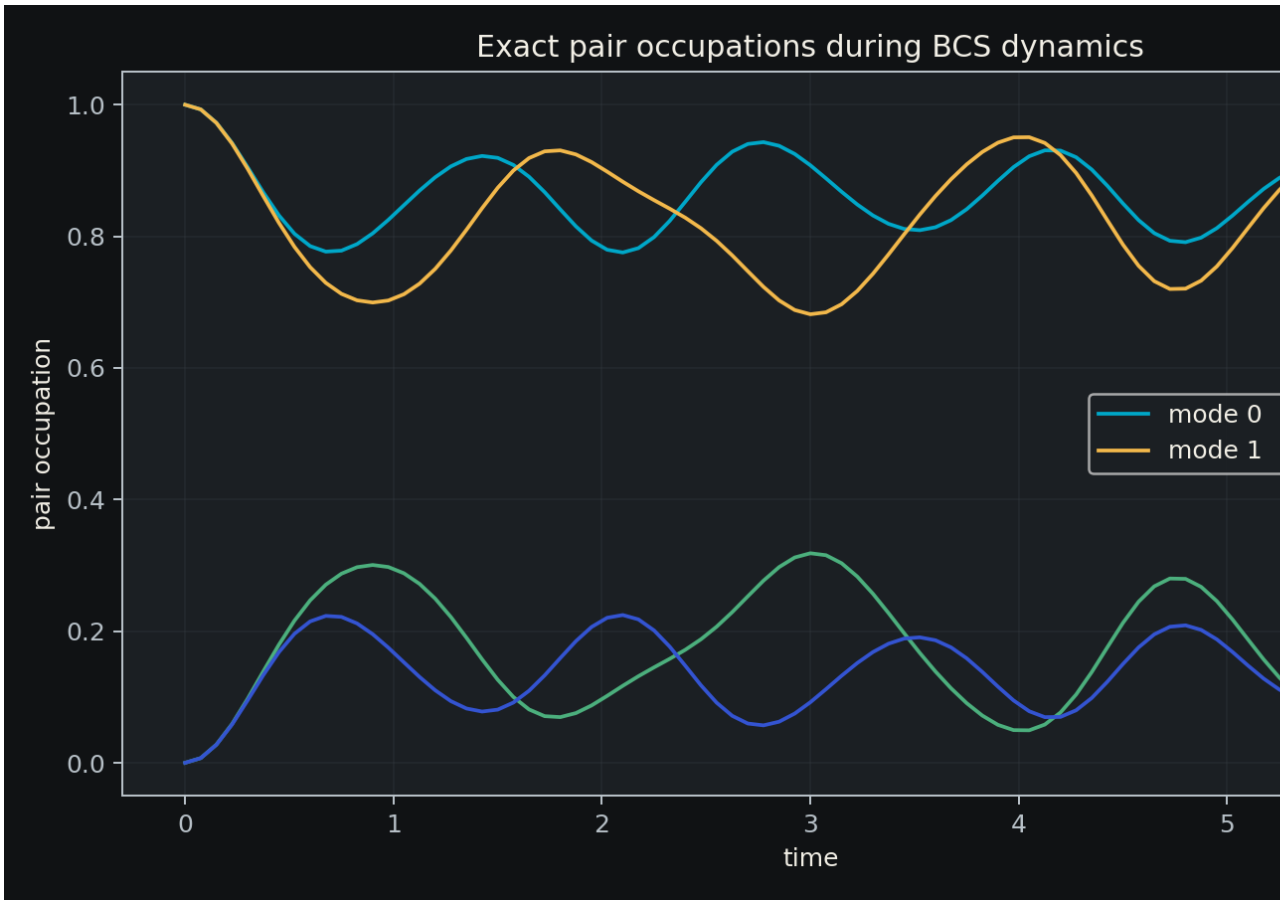
```
time                 6.000000
fidelity             1.000000
state_error          0.001410
exact_pair_number    2.000000
trotter_pair_number  2.000000
```

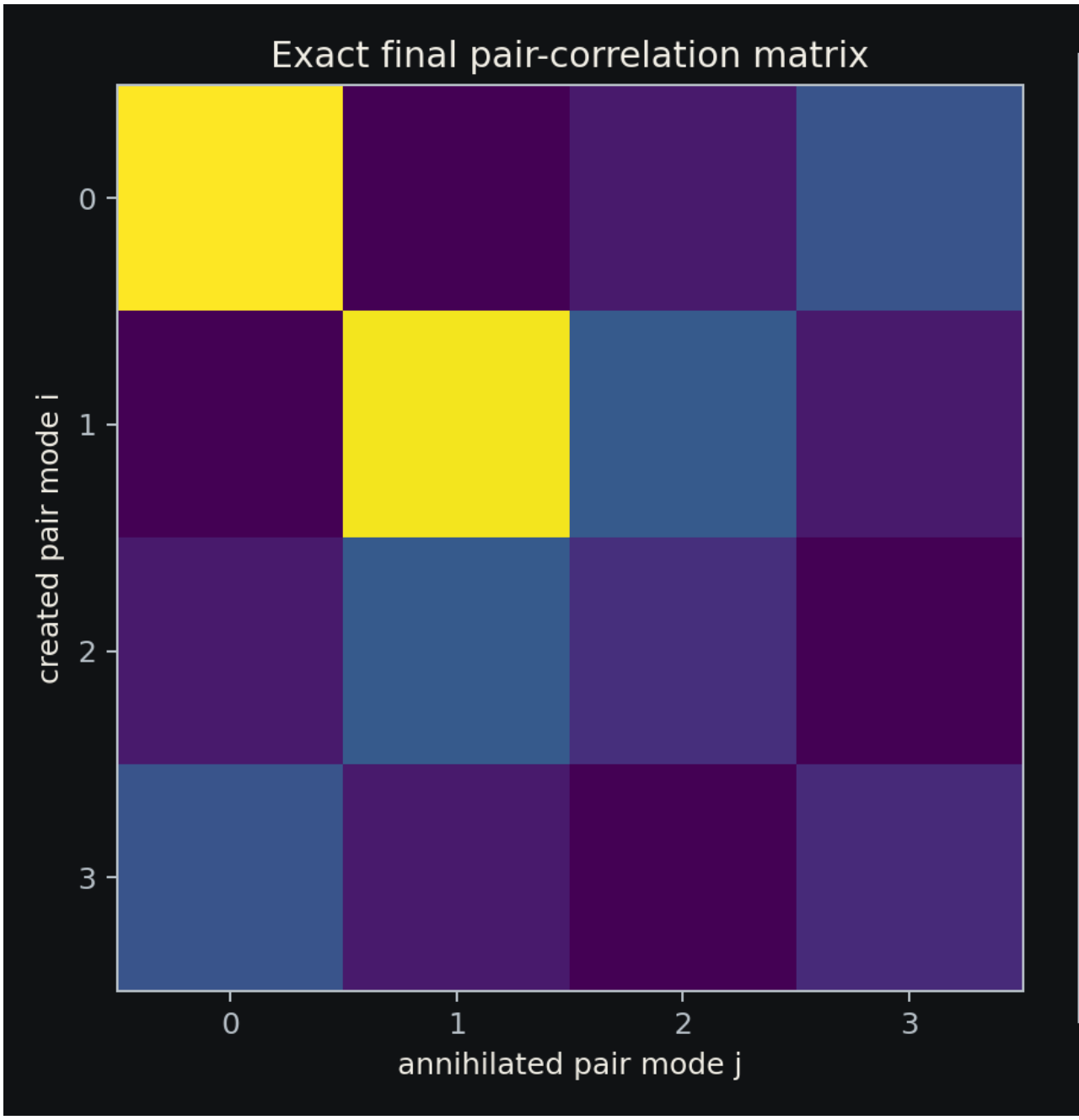
```
exact_gap_proxy        0.826271
trotter_gap_proxy      0.826182
Name: 80, dtype: float64
```

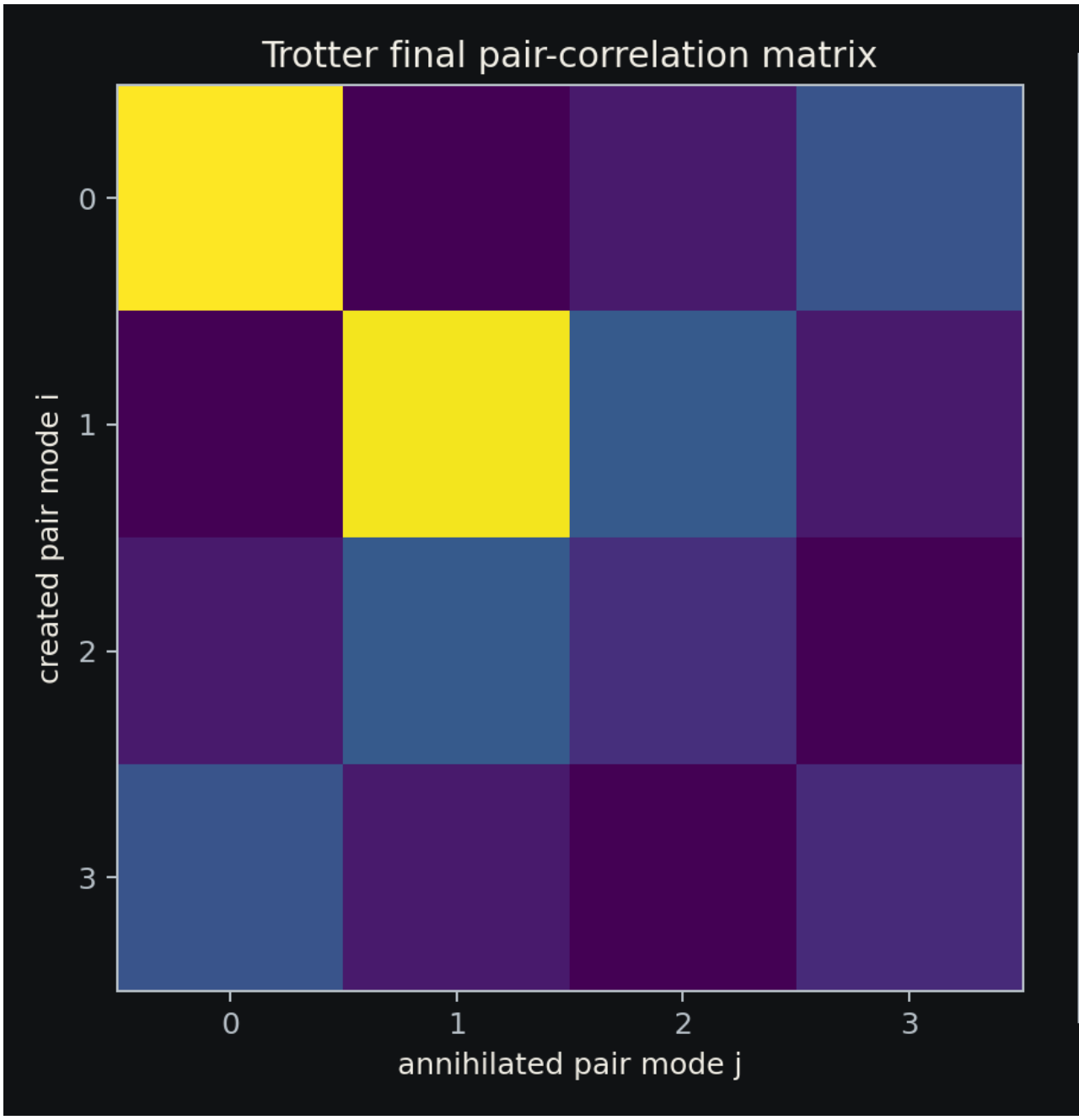
In [7]:

```
for fig in [
    'bcs_gap_condensate_time.png',
    'bcs_pair_occupations_time.png',
    'bcs_pair_correlation_exact_final.png',
    'bcs_pair_correlation_trotter_final.png',
]:
    show_figure(fig, width=760)
```


Reduced BCS observables for paper figures
effective gap proxy
0.92
0.90
0.88
0.86
0.84
0.82
0.80
0.78
largest pair-correlation eigenvalue
1.30
1.25
1.20
1.15
1.10
1.05
1.00
0.95
0
1
2
3
4
5
time


Exact pair occupations during BCS dynamics
pair occupation
time
mode 0
mode 1
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5


Exact final pair-correlation matrix
created pair mode i
annihilated pair mode j
0
1
2
3
0
1
2
3



Green normal, Gor'kov anômalo e matriz Nambu¶

In [8]:

```
green_spec = read_table('bcs_pair_green_spectral_density.csv')
gorkov_spec = read_table('bcs_gorkov_anomalous_spectrum.csv')
nambu_spec = read_table('bcs_gorkov_nambu_spectral_trace.csv')

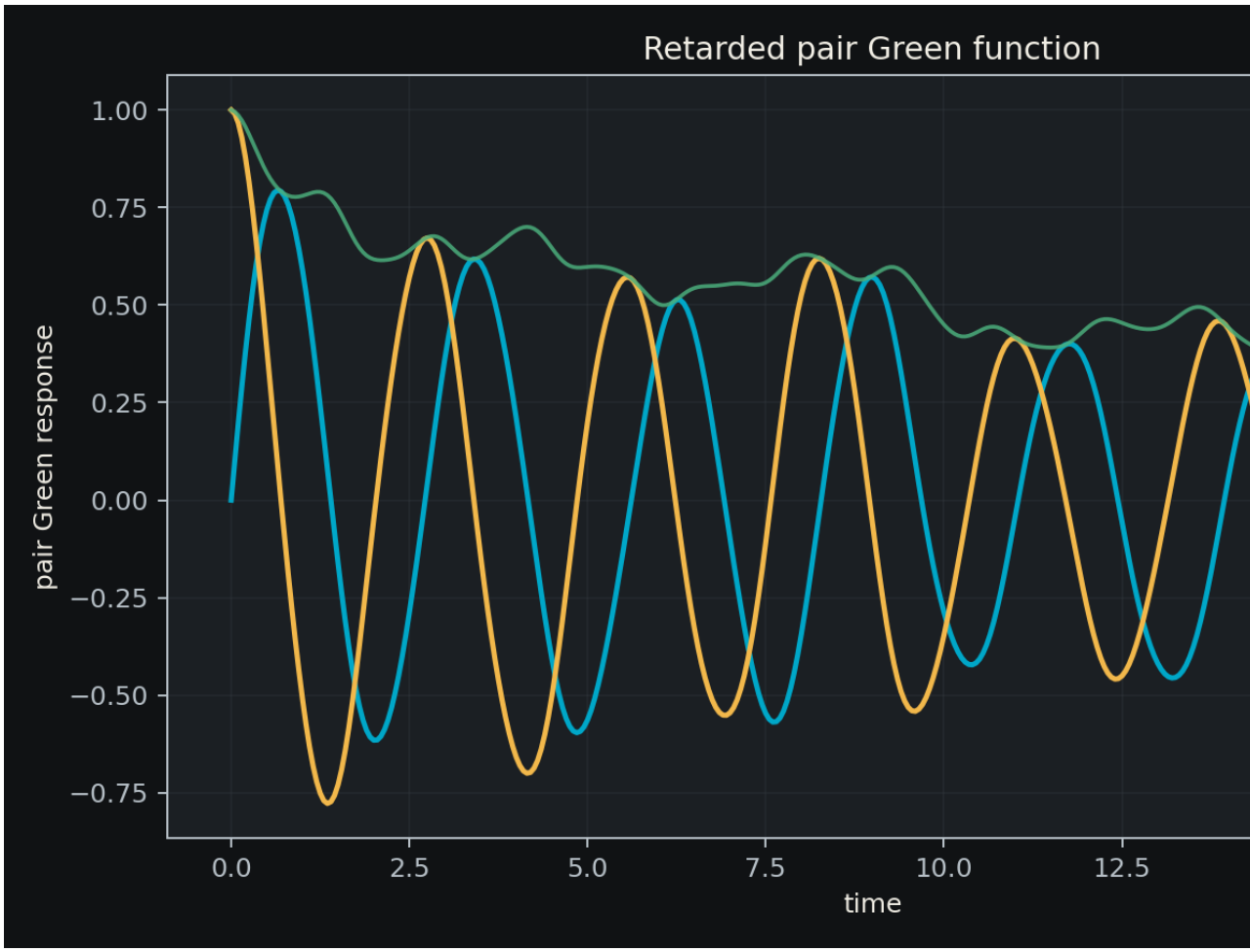
summary_spectral = pd.DataFrame([
    {'channel': 'normal G', **green_spec.sort_values('spectral_weight', ascending=False).ilo
    {'channel': 'anomalous F', **gorkov_spec.sort_values('spectral_weight', ascending=False)
    {'channel': 'Nambu trace', **nambu_spec.sort_values('spectral_weight', ascending=False).
])
summary_spectral
```

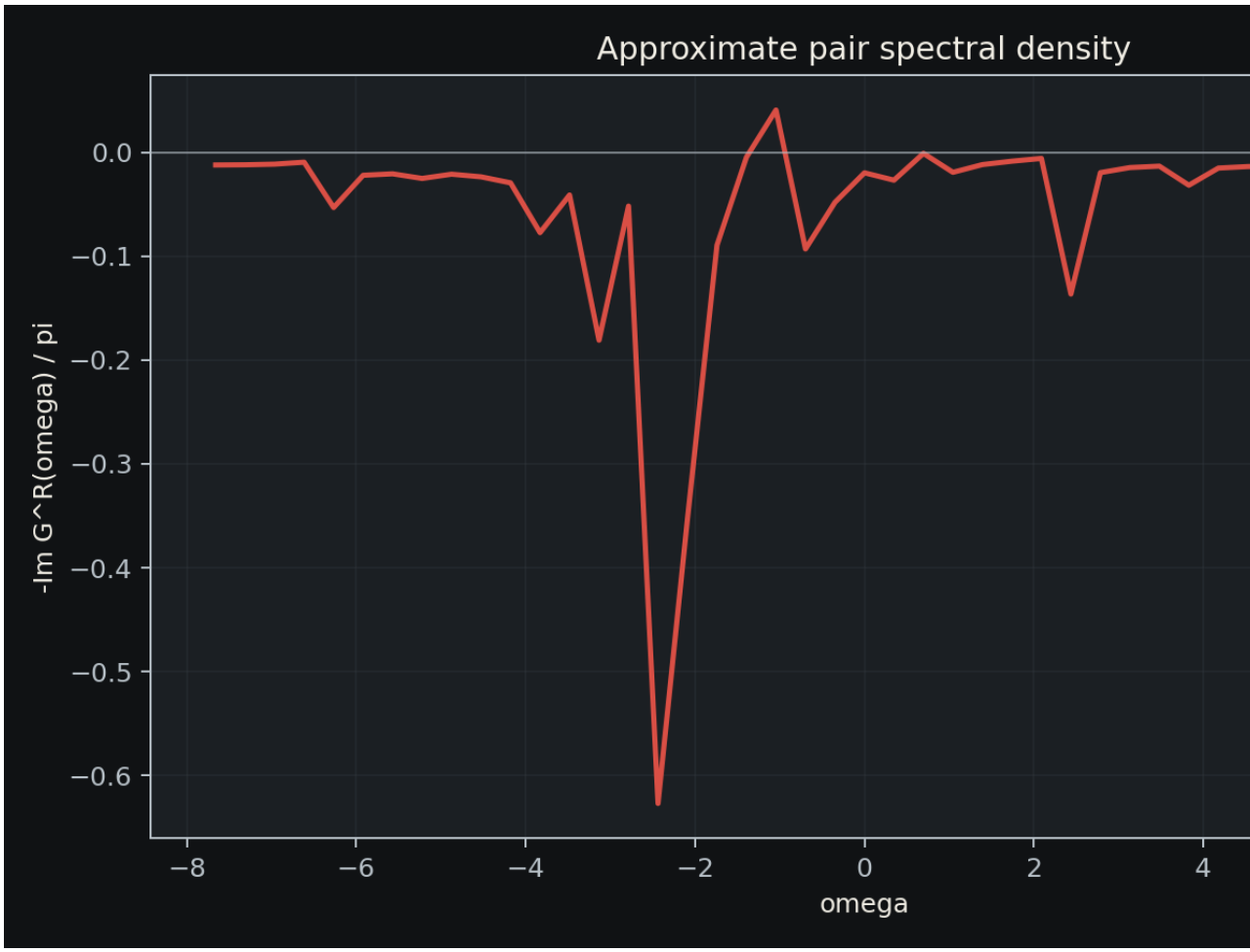
Out[8]:

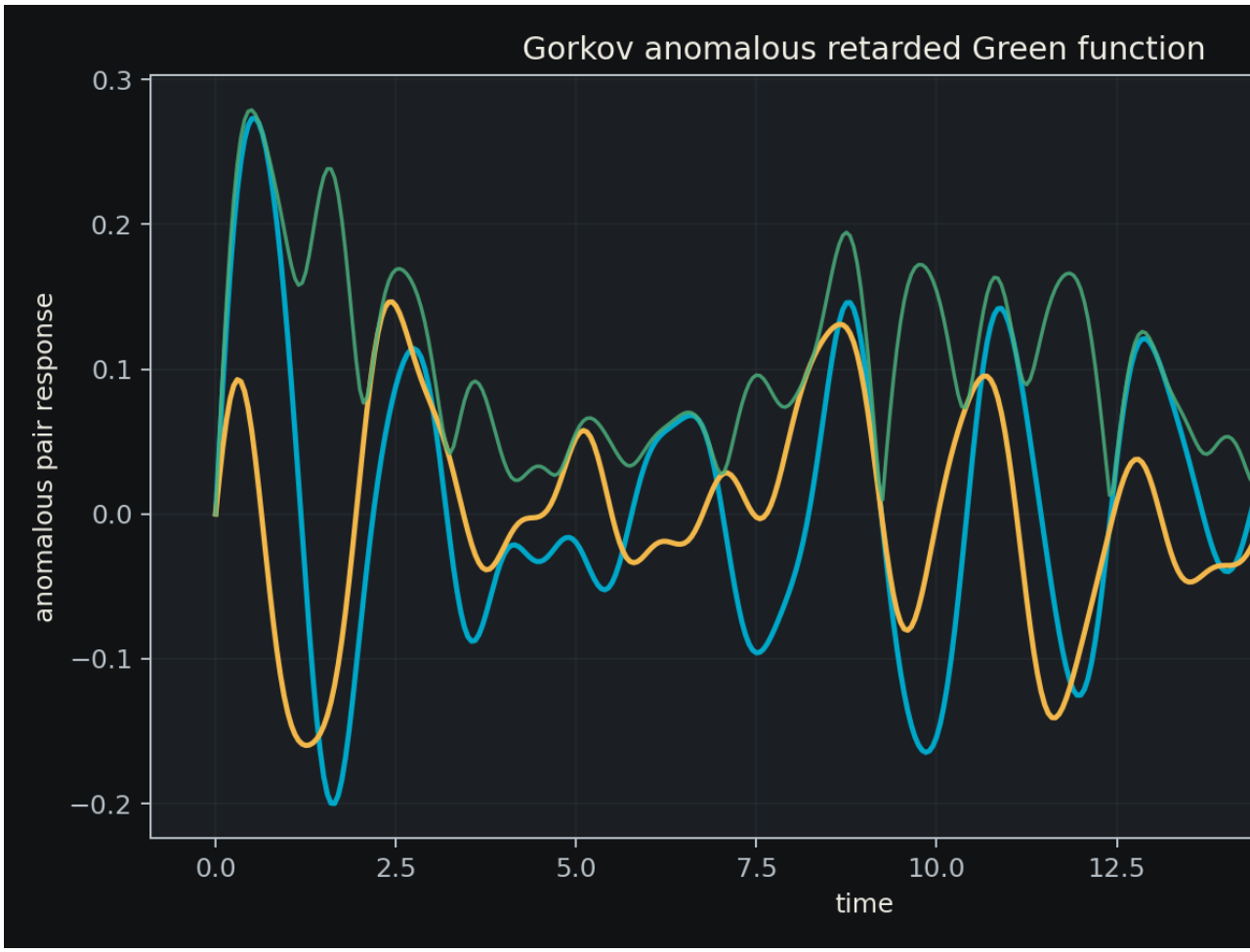
	channel	omega	spectral_weight
0	normal G	-1.044297	0.040784
1	anomalous F	3.132890	0.232452
2	Nambu trace	-2.436692	0.673764

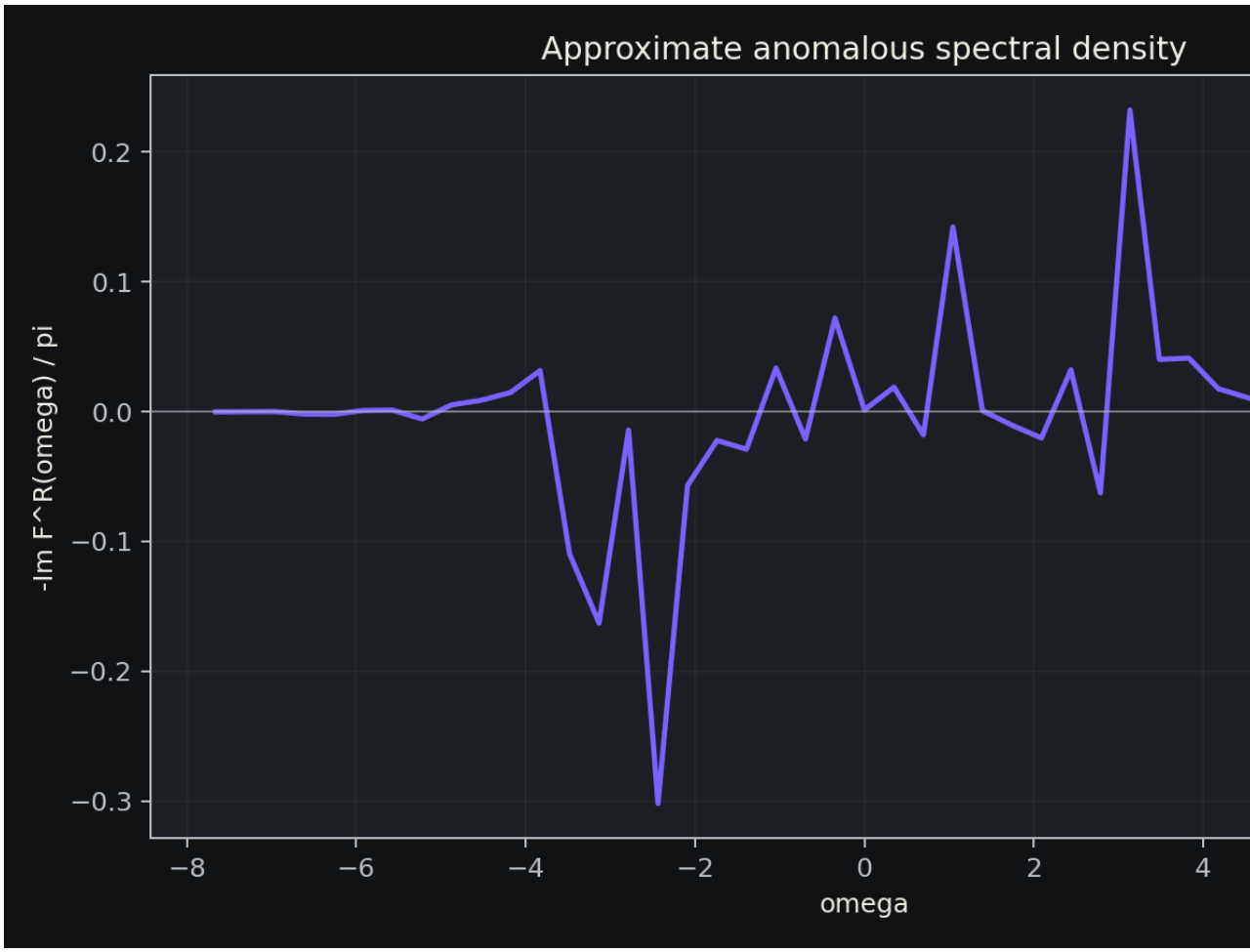
In [9]:

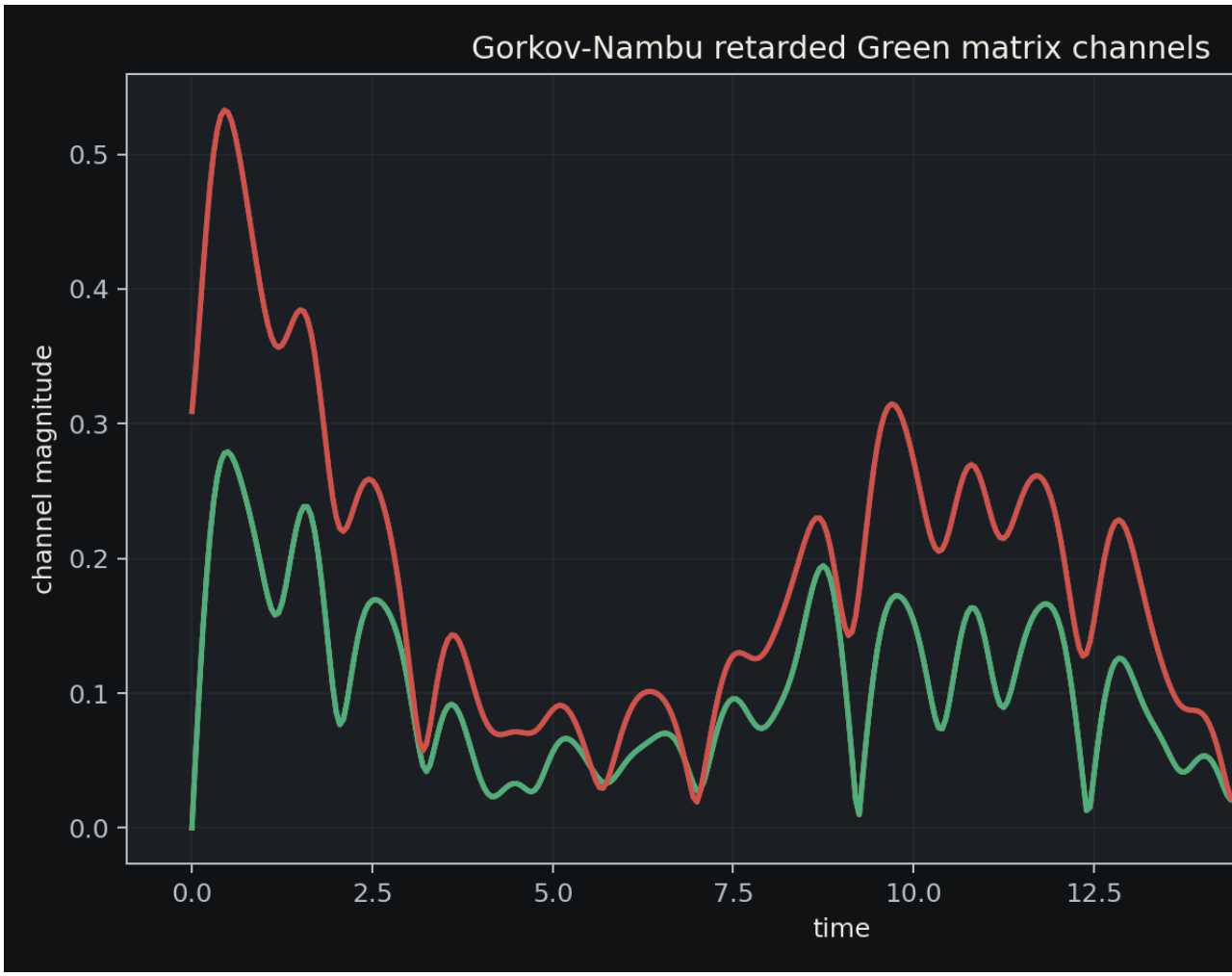
```
for fig in [
    'bcs_pair_green_time.png',
    'bcs_pair_green_spectrum.png',
    'bcs_gorkov_anomalous_time.png',
    'bcs_gorkov_anomalous_spectrum.png',
    'bcs_gorkov_nambu_channels.png',
    'bcs_gorkov_nambu_spectral_trace.png',
]:
    show_figure(fig, width=820)
```

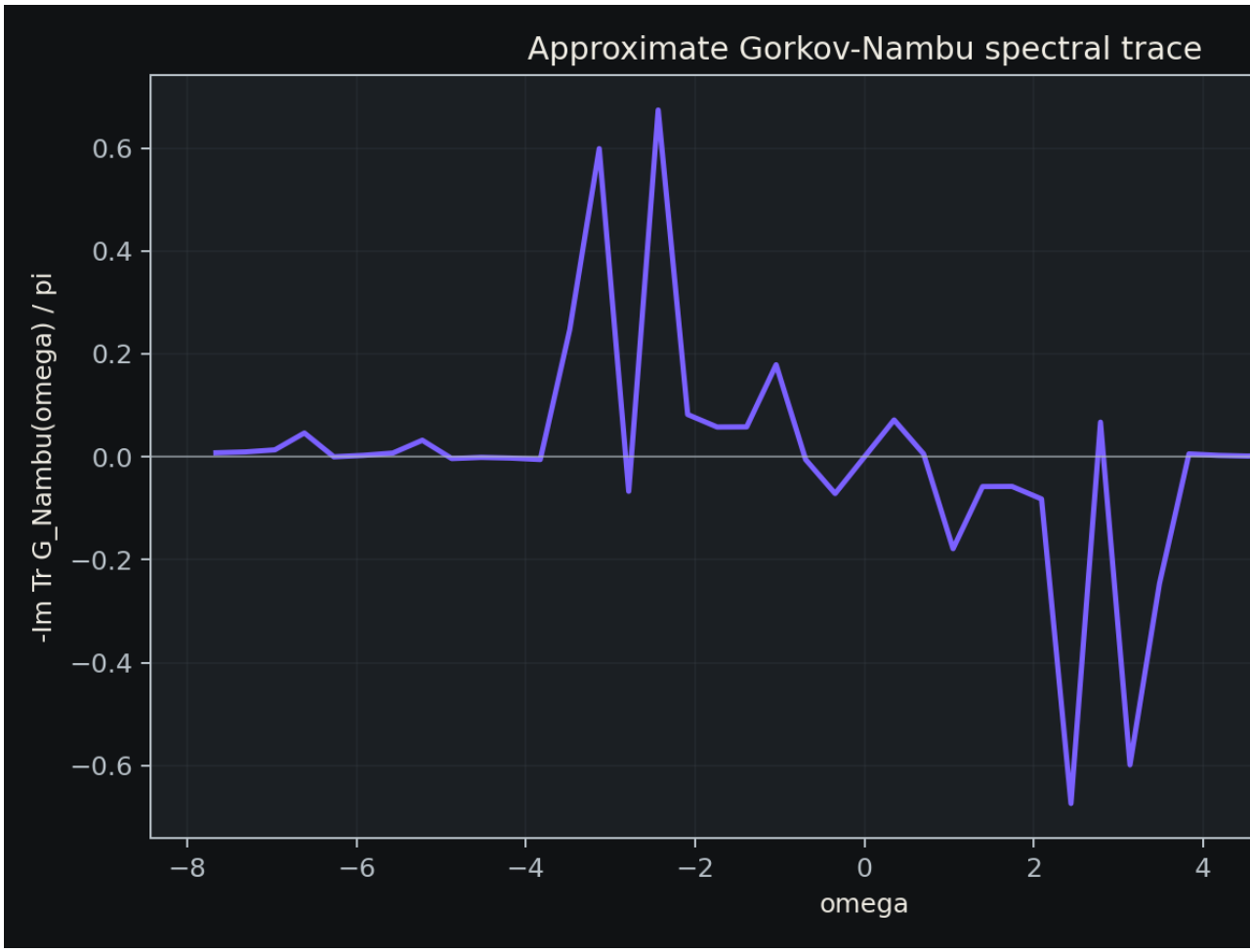

Retarded pair Green function
pair Green response
time
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0.0
2.5
5.0
7.5
10.0
12.5


Approximate pair spectral density
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
-Im G^R(omega) / pi
−8
−6
−4
−2
0
2
4
omega


Gorkov anomalous retarded Green function
anomalous pair response
time
0.3
0.2
0.1
0.0
−0.1
−0.2
0.0
2.5
5.0
7.5
10.0
12.5


Approximate anomalous spectral density
-Im F^R(omega) / pi
0.2
0.1
0.0
−0.1
−0.2
−0.3
−8
−6
−4
−2
0
2
4
omega


Gorkov-Nambu retarded Green matrix channels
channel magnitude
0.0
0.1
0.2
0.3
0.4
0.5
0.0
2.5
5.0
7.5
10.0
12.5
time



Qiskit shots, energia por Pauli e mitigação¶

In [10]:

```
shots = read_table('bcs_qiskit_shots_noise.csv')
energy = read_table('bcs_qiskit_energy_shots.csv')
mitigation = read_table('bcs_qiskit_error_mitigation.csv')

display(shots[['mode', 'shots', 'total_pair_number', 'total_pair_error']])
display(energy[['mode', 'shots', 'exact_energy', 'statevector_energy', 'energy', 'abs_error_
display(mitigation[['mode', 'noise_scale', 'energy', 'abs_error_vs_statevector', 'noise_slop
```

	mode	shots	total_pair_number	total_pair_error
0	ideal_shots	1024.0	2.000000	3.330669e-14

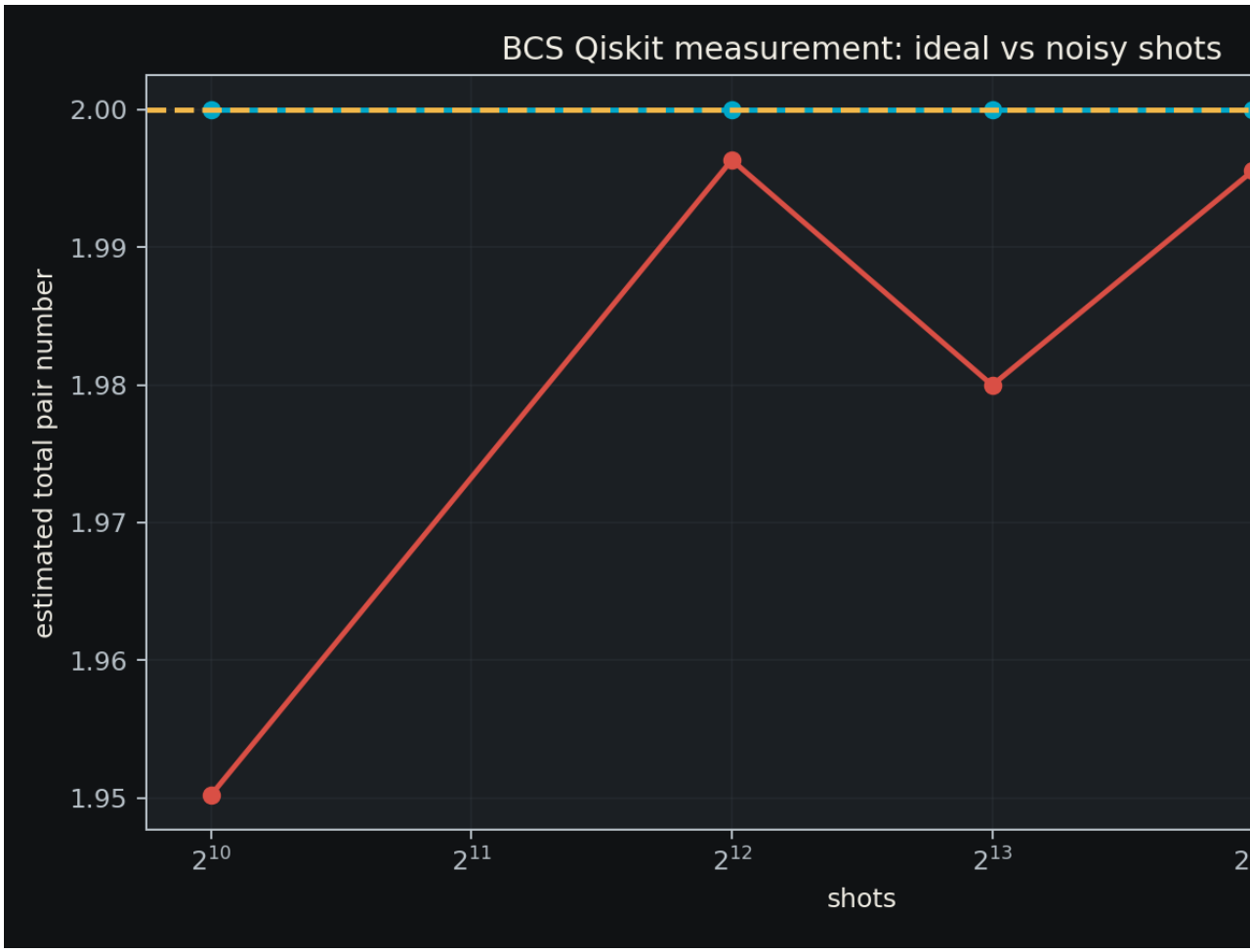
	mode	shots	total_pair_number	total_pair_error
1	noisy_shots	1024.0	1.950195	4.980469e-02
2	ideal_shots	4096.0	2.000000	3.330669e-14
3	noisy_shots	4096.0	1.996338	3.662109e-03
4	ideal_shots	8192.0	2.000000	3.330669e-14
5	noisy_shots	8192.0	1.979980	2.001953e-02
6	ideal_shots	16384.0	2.000000	3.330669e-14
7	noisy_shots	16384.0	1.995544	4.455566e-03
8	ideal_shots	32768.0	2.000000	3.330669e-14
9	noisy_shots	32768.0	1.989685	1.031494e-02

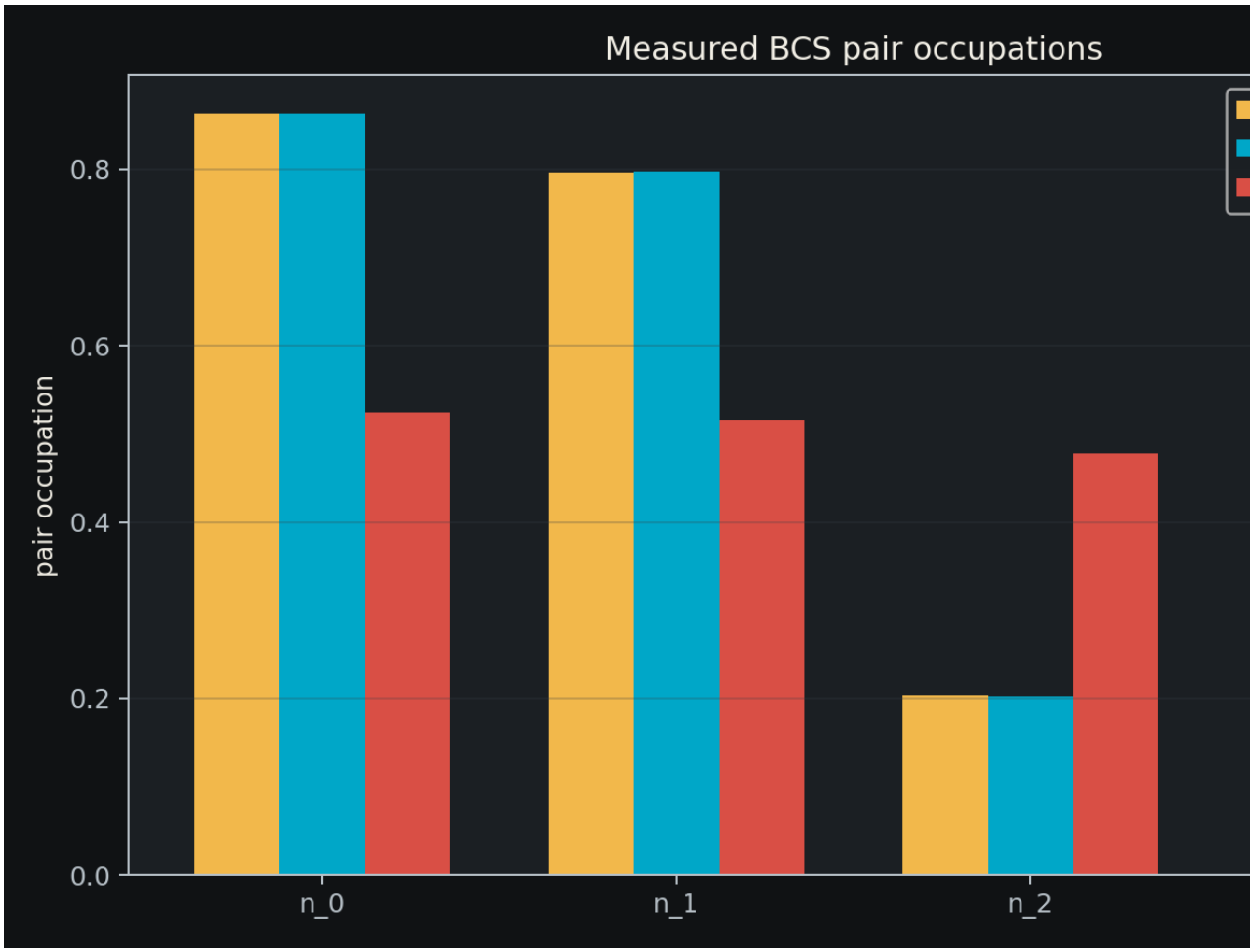
	mode	shots	exact_energy	statevector_energy	energy	abs_error_vs_statevector
0	ideal_shots	1024.0	2.7	2.61166	2.625391	0.013731
1	noisy_shots	1024.0	2.7	2.61166	0.125879	2.485781
2	ideal_shots	4096.0	2.7	2.61166	2.652319	0.040659
3	noisy_shots	4096.0	2.7	2.61166	0.244165	2.367495
4	ideal_shots	8192.0	2.7	2.61166	2.605103	0.006558
5	noisy_shots	8192.0	2.7	2.61166	0.210583	2.401077
6	ideal_shots	16384.0	2.7	2.61166	2.607983	0.003677
7	noisy_shots	16384.0	2.7	2.61166	0.202246	2.409414
8	ideal_shots	32768.0	2.7	2.61166	2.611313	0.000347
9	noisy_shots	32768.0	2.7	2.61166	0.214069	2.397592

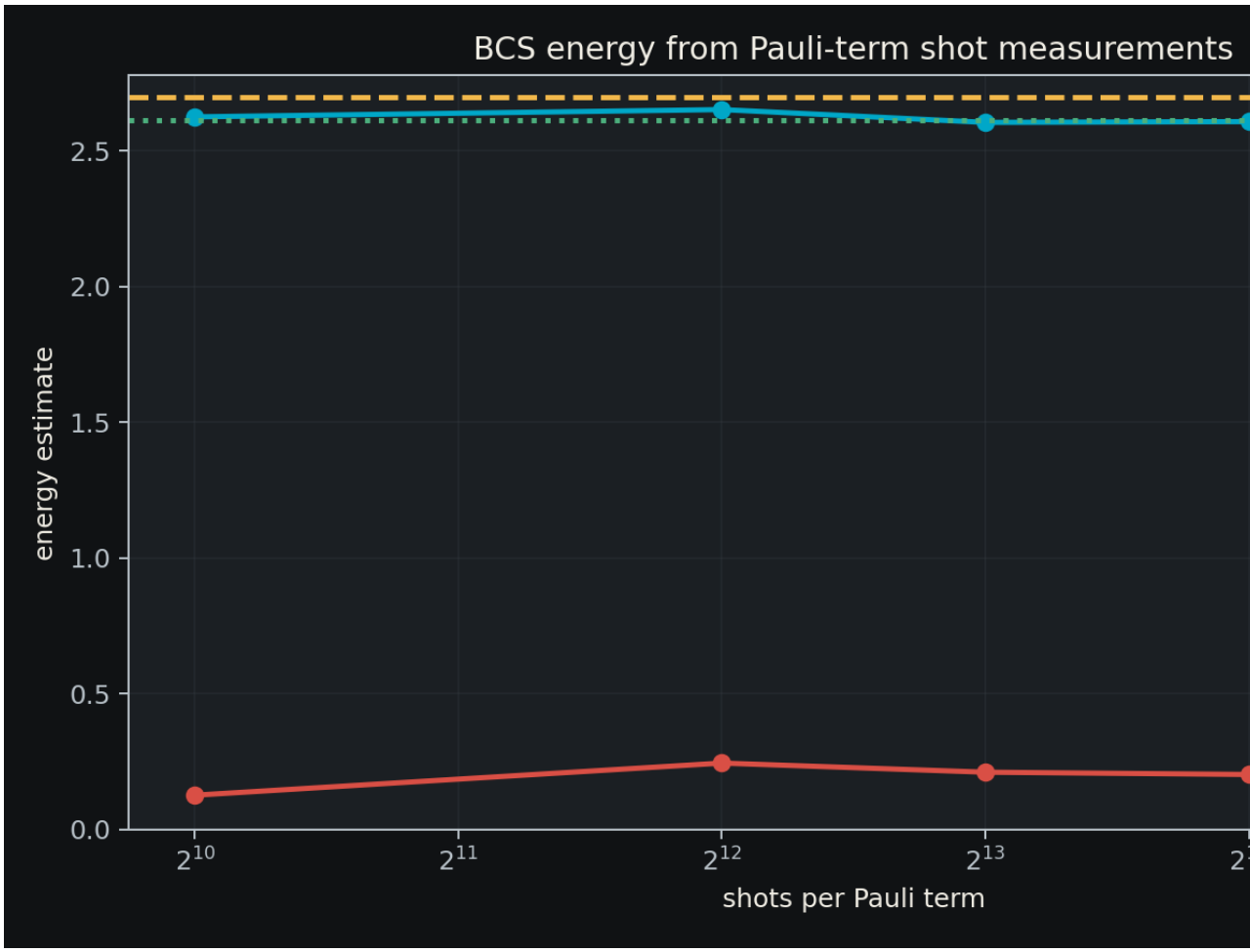
	mode	noise_scale	energy	abs_error_vs_statevector	noise_slope	fit_rmse
0	scaled_noise	0.25	2.219580	0.392080	NaN	NaN
1	scaled_noise	0.50	1.915979	0.695681	NaN	NaN
2	scaled_noise	0.75	1.632208	0.979452	NaN	NaN
3	scaled_noise	1.00	1.404230	1.207430	NaN	NaN
4	linear_zne	0.00	2.475455	0.136205	-1.091929	0.019328

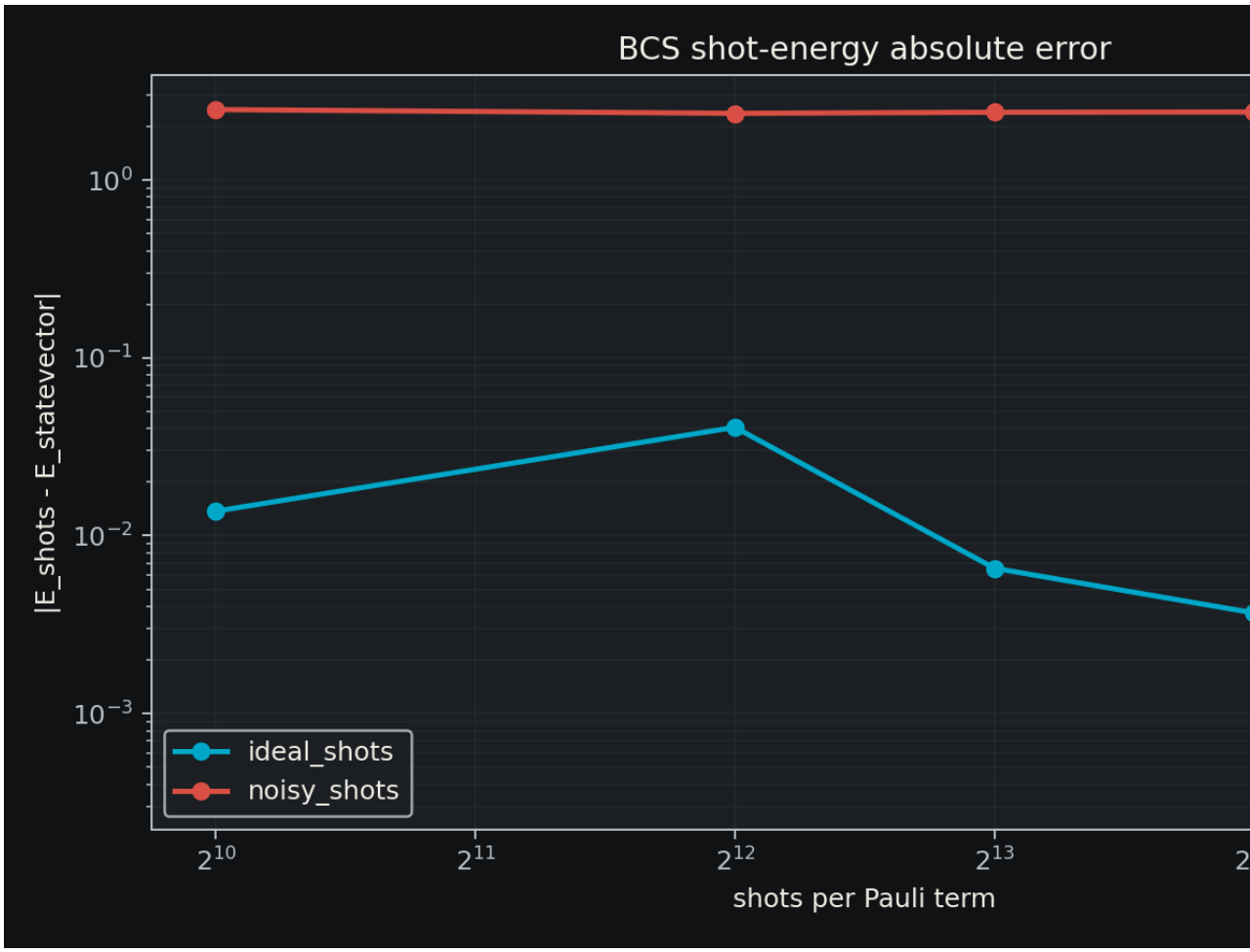
In [11]:

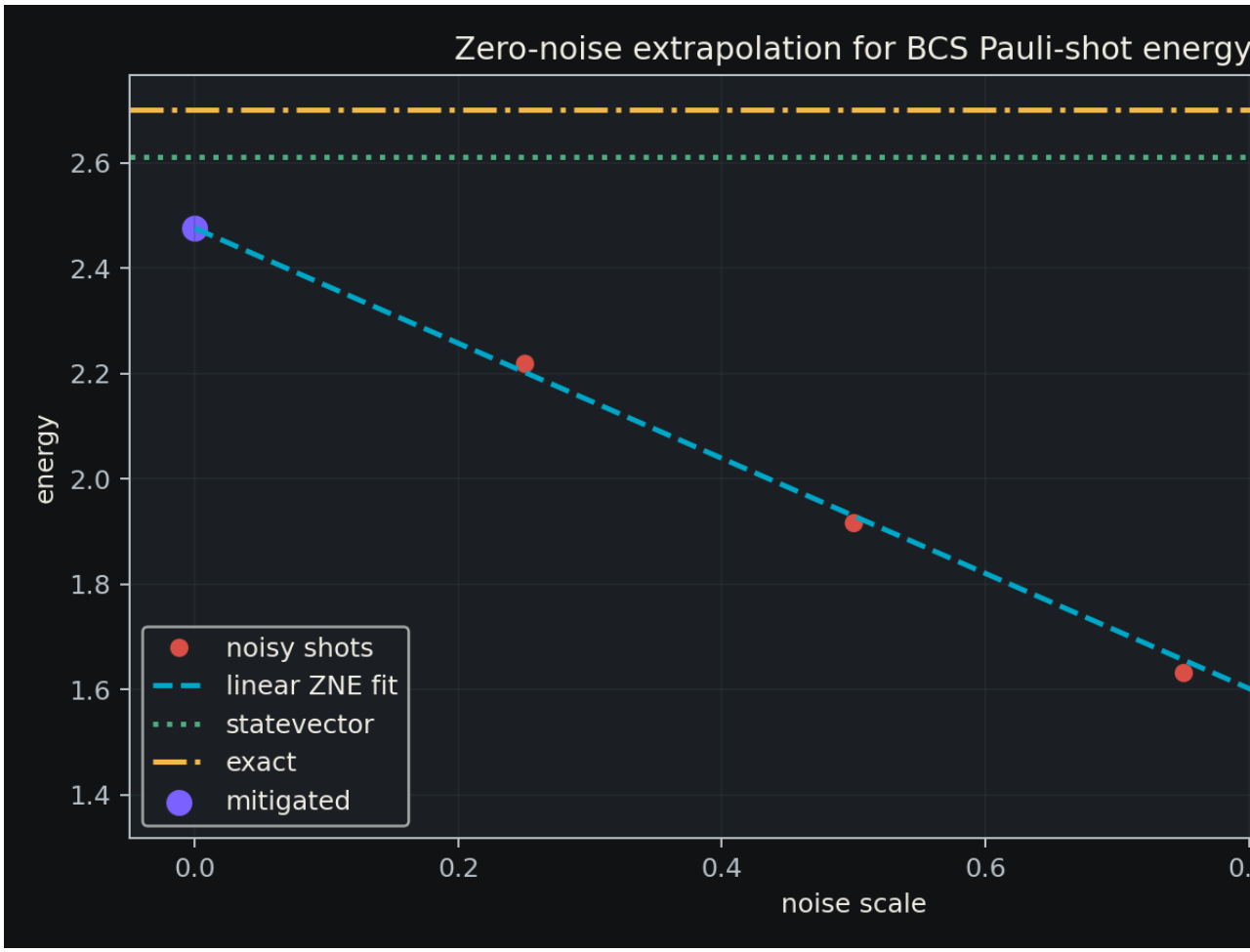
```
for fig in [
    'bcs_qiskit_shots_noise_total_pairs.png',
    'bcs_qiskit_shots_noise_occupations.png',
    'bcs_qiskit_energy_shots_values.png',
    'bcs_qiskit_energy_shots_error.png',
    'bcs_qiskit_error_mitigation_zne.png',
    'bcs_qiskit_error_mitigation_errors.png',
]:
    show_figure(fig, width=820)
```

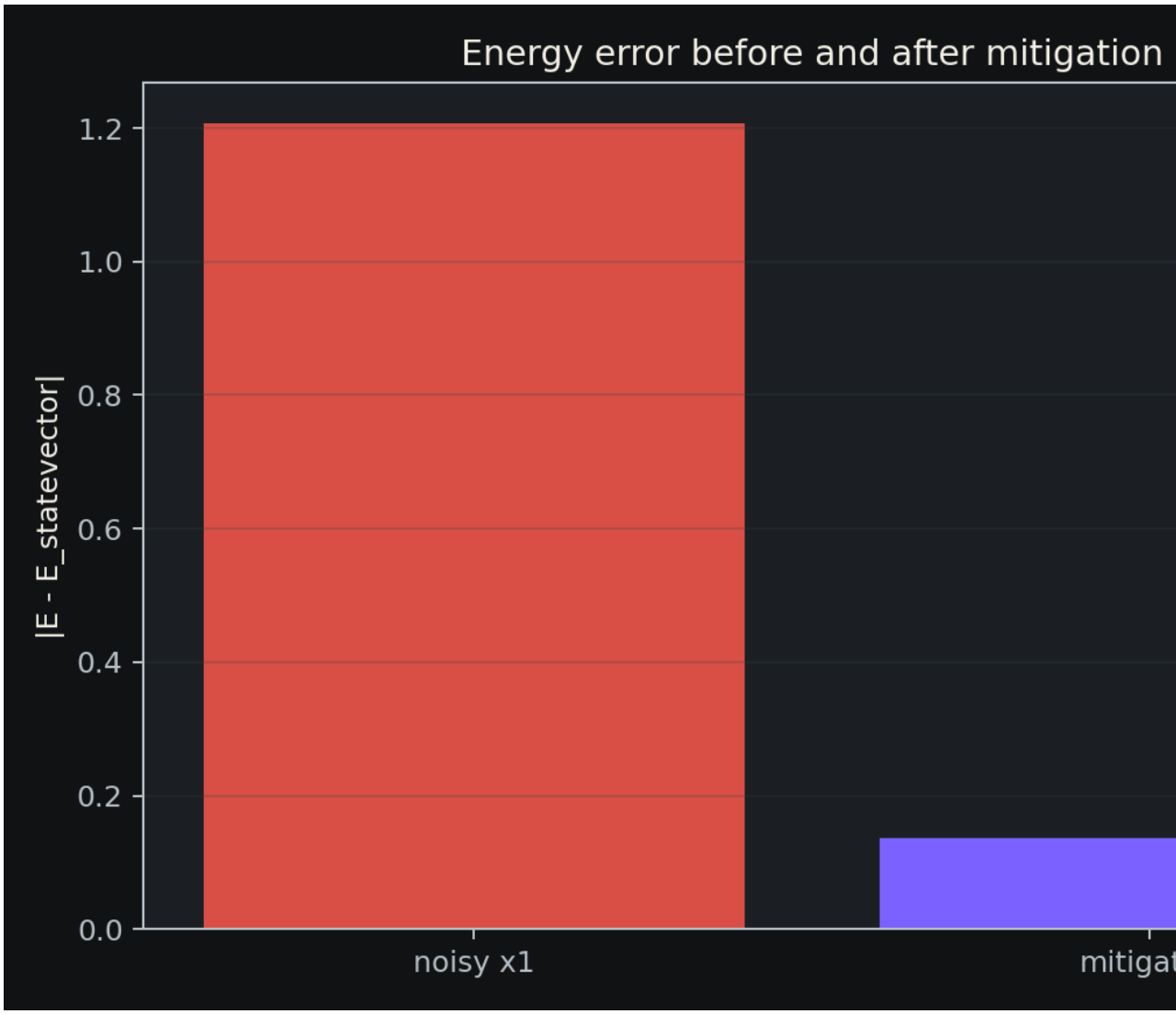

BCS Qiskit measurement: ideal vs noisy shots
estimated total pair number
2.00
1.99
1.98
1.97
1.96
1.95
2^{10}
2^{11}
2^{12}
2^{13}
shots


Measured BCS pair occupations
pair occupation
0.0
0.2
0.4
0.6
0.8
n_0
n_1
n_2


BCS energy from Pauli-term shot measurements
energy estimate
2.5
2.0
1.5
1.0
0.5
0.0
2^{10}
2^{11}
2^{12}
2^{13}
shots per Pauli term


BCS shot-energy absolute error
|E_shots - E_statevector|
10^{0}
10^{-1}
10^{-2}
10^{-3}
ideal_shots
noisy_shots
2^{10}
2^{11}
2^{12}
2^{13}
shots per Pauli term


Zero-noise extrapolation for BCS Pauli-shot energy
energy
2.6
2.4
2.2
2.0
1.8
1.6
1.4
noisy shots
linear ZNE fit
statevector
exact
mitigated
0.0
0.2
0.4
0.6
noise scale



Resumo numérico final¶

In [12]:

```
energy_32768 = energy[(energy['mode'] == 'ideal_shots') & (energy['shots'] == 32768)].iloc[0
noisy_scale_1 = mitigation[(mitigation['mode'] == 'scaled_noise') & (mitigation['noise_scale
zne = mitigation[mitigation['mode'] == 'linear_zne'].iloc[0]

paper_summary = pd.DataFrame([
    {'metric': 'final observable fidelity', 'value': final_obs['fidelity']},
    {'metric': 'final state error', 'value': final_obs['state_error']},
```

```
    {'metric': 'normal Green peak omega', 'value': summary_spectral.loc[0, 'omega']},
    {'metric': 'anomalous Green peak omega', 'value': summary_spectral.loc[1, 'omega']},
    {'metric': 'Nambu trace peak omega', 'value': summary_spectral.loc[2, 'omega']},
    {'metric': '32768 ideal shot energy error vs statevector', 'value': energy_32768['abs_er
    {'metric': 'noisy energy error before mitigation', 'value': noisy_scale_1['abs_error_vs_
    {'metric': 'energy error after ZNE mitigation', 'value': zne['abs_error_vs_statevector']
])
paper_summary
```

Out[12]:

	metric	value
0	final observable fidelity	1.000000
1	final state error	0.001410
2	normal Green peak omega	-1.044297
3	anomalous Green peak omega	3.132890
4	Nambu trace peak omega	-2.436692
5	32768 ideal shot energy error vs statevector	0.000347
6	noisy energy error before mitigation	1.207430
7	energy error after ZNE mitigation	0.136205

Regenerar resultados¶

Execute no terminal, a partir da raiz do repositório:

```
source .venv/bin/activate
python experiments/run_all.py
pandoc papers/bcs_quantum_simulation.md -s -o papers/bcs_quantum_simulation.html
google-chrome --headless --disable-gpu --no-sandbox \
  --print-to-pdf="$PWD/papers/bcs_quantum_simulation.pdf" \
  "file://$PWD/papers/bcs_quantum_simulation.html"
```

Para validar o núcleo:

```
PYTHONPATH=src .venv/bin/python -m pytest -q tests
```